

# Factor a difference of two squares

Algebra I | CK-12: Factoring Special Products

## Key Idea

A **difference of squares**  $a^2 - b^2$  always factors as  $(a + b)(a - b)$ . Both terms must be perfect squares with a minus between them.

## Key Terms

- **Perfect square** — a number/expression that is something squared
- **Difference of squares** —  $a^2 - b^2$

## Worked steps

Step 1: Check both terms are perfect squares and separated by a minus.

Step 2: Write  $(a + b)(a - b)$  where  $a, b$  are the square roots.

Example:  $x^2 - 9 = (x + 3)(x - 3)$ .

## You Try

Factor  $x^2 - 25$ .